Fintech Course Group Project 2

Data-Driven Investment Needs Estimation and Personalized Recommendation System

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Abstract

This study develops an end-to-end KYC (Know Your Client) pipeline to estimate investment needs for wealth management clients and deliver MiFID II-compliant personalized product recommendations. Using a revealed-preference labeling scheme on 5,000 anonymized clients, we frame the problem as calibrated binary classification across two investment need types: Accumulation (% positive rate) and Income (% positive rate). Exploratory analysis reveals that the two targets are empirically near-independent ($r =$), that Wealth and Income require logarithmic transformation, and that FinancialEducation and RiskPropensity are multicollinear ($r =$). A structured ablation against the professor's baseline feature set demonstrates that our theoretically motivated engineered feature set lifts linear models significantly but is redundant for tree-based models, which learn interactions natively. Per-model preprocessing pipelines are matched to each algorithm's inductive bias. Model selection uses nested cross-validation (outer 10-fold for selection, inner 3-fold for hyperparameter tuning) with a Wilcoxon signed-rank test between top candidates, F1 as the primary metric, and Precision ≥ 0.75 as a hard MiFID II compliance constraint. Operating thresholds are chosen from the Precision–Recall curve rather than fixed at 0.5. The winning Soft Voting Ensemble generates calibrated propensity scores that feed a three-stage recommendation engine covering both need types, ranking clients by confidence-weighted suitability under a hard regulatory risk cap.

Keywords: *Investment Needs, KYC, Propensity Scoring, MiFID II, XGBoost, Ensemble Methods, Feature Engineering, Calibration, Nested CV*

1. INTRODUCTION

Wealth management firms are legally required under MiFID II to match investment products to demonstrable client needs. Identifying those needs at scale through manual advisory processes is operationally infeasible and introduces advisor-specific biases that harm both compliance and client outcomes. Data-driven KYC systems address this by learning need propensities directly from client attributes and behavioral history.

In this work we build such a system end-to-end, grounding every modeling decision in a combination of statistical evidence from the data and theoretical prior from financial economics. The regulatory context is not a post-hoc constraint but the organizing principle of the entire pipeline: the choice of evaluation metric, the precision floor, the confidence-based segmentation of clients, and the hard risk cap in product matching all derive from MiFID II requirements.

The dataset contains 5,000 anonymized clients of a large wealth management firm. Two binary response variables encode investment needs: *AccumulationInvestment* (systematic dollar-cost averaging, characteristic of working-age clients in the capital accumulation phase) and *IncomeInvestment* (lump-sum deployment of existing capital for cash flow, characteristic of clients approaching or in retirement). Labels derive from a *revealed preference* scheme: if a trusted advisor sold a product matching a given need type and the client purchased it, we infer the client held that need. The model learns to clone the behavior of reliable advisors.

A companion dataset lists 11 investment products with their type (Accumulation or Income) and a normalized Synthetic Risk Indicator (SRI) in $[0.12, 0.88]$, which feeds the product matching stage.

2. EXPLORATORY ANALYSIS

2.1. Distributional Properties

Table 1 summarizes key descriptive statistics of the Needs dataset.

Wealth and Income are severely right-skewed. Wealth has mean 93.8k but standard deviation 105.5k and maximum 2,233k, indicating a distribution closer to a power law than a Gaussian. Income exhibits analogous behavior. Both variables violate the approximate symmetry assumed by distance-based and gradient-based models. We apply `log1p` to both, yielding distributions that are approximately Gaussian and substantially better conditioned. Box-plots confirm extreme outliers in raw scale that are absorbed into the bulk of the distribution after transformation.

Table 1. Needs dataset: selected variable statistics ($n = 5{,}000$).

Variable	Mean	Std	Min	Max
Age	55.25	11.97	18.00	97.00
Gender (F = 1)	0.49	0.50	0.00	1.00
FamilyMembers	2.51	0.76	1.00	5.00
FinancialEducation	0.42	0.15	0.04	0.90
RiskPropensity	0.36	0.15	0.02	0.88
Income (k)	62.99	44.36	1.54	365.32
Wealth (k)	93.81	105.47	1.06	2233.23
IncomeInvestment	0.38	0.49	0.00	1.00
AccumulationInvestment	0.51	0.50	0.00	1.00

RiskPropensity and FinancialEducation are already normalized to $[0, 1]$ with no missing values, consistent with their origin as MiFID profile scores. Their mutual correlation of $r = 0.68$ demands careful treatment in regularized linear models (Section 2.2).

Target class balance differs. AccumulationInvestment has a 51% positive rate — effectively balanced, meaning standard metrics apply without correction. IncomeInvestment has a 38% positive rate — mild but non-trivial imbalance. We apply `scale_pos_weight` in XGBoost and `class_weight='balanced'` in Logistic Regression and SVM for IncomeInvestment only. Applying these corrections to the balanced AccumulationInvestment target would degrade performance by over-weighting a non-problem. SMOTE is deliberately excluded: it is designed for severe imbalance (e.g. 5/95 ratios) and introducing synthetic observations at a 38% minority rate would add noise without measurable benefit.

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Investment Needs Classification and Next Best Action

2.2. Correlation Structure

The two targets are empirically near-independent. The correlation between IncomeInvestment and AccumulationInvestment is $r = 0.011$, essentially zero. This is a critical empirical correction to the prior theoretical expectation of mutual exclusivity implied by the Modigliani life-cycle hypothesis. Clients can hold both needs simultaneously or neither. Consequently, (1) independent binary classifiers are statistically justified; (2) the Classifier Chain approach offers limited empirical gain from inter-target information transfer, though it encodes a structural financial prior and is tested empirically.

IncomeInvestment has a stronger signal. Its strongest marginal correlations are Age ($r = 0.33$) and Wealth_log ($r = 0.40$), consistent with the life-cycle hypothesis. For AccumulationInvestment the strongest predictor is Income_log ($r = 0.32$), followed by Wealth_log ($r = 0.12$) and RiskPropensity ($r = 0.068$). The weaker signal structure means AccumulationInvestment will yield more variable model performance.

FinancialEducation and RiskPropensity are collinear ($r = 0.68$). In Logistic Regression this inflates coefficient variance. We address it via L1 regularization (which zeros out one variable automatically) and by engineering the interaction term FinEdu_x_Risk = FinancialEducation × RiskPropensity, giving LR an explicit cross-product it cannot reconstruct from correlated main effects alone.

Wealth and Income are correlated ($r = 0.60$ raw, $r = 0.67$ log-transformed). Moderate and economically expected; not a multicollinearity problem for tree-based models.

2.3. Visual Motivation for Engineered Features

Violin plots — RiskPropensity by age group. Among clients aged 65+, those with IncomeInvestment = 1 show *higher* risk propensity than those with IncomeInvestment = 0, reversing the pattern seen in younger cohorts. This is consistent with the life-cycle hypothesis: wealthier elderly clients can afford more risk while still seeking income. For AccumulationInvestment, age group has virtually no effect — distribution shapes are nearly identical across all brackets, confirming the weaker signal structure identified in the correlation matrix.

Scatter plot — Age × Wealth. Clients with IncomeInvestment = 1 cluster in the top-right quadrant (old *and* wealthy), confirming that the joint effect of age and wealth is stronger than either marginal predictor alone. This is the direct visual motivation for the `Age_x_Wealth` interaction feature.

Scatter plot — Income × Wealth. Class separation for AccumulationInvestment follows the horizontal axis (Income) rather than the vertical (Wealth), confirming that current income — not accumulated capital — is the primary driver of accumulation behavior. This motivates the per-capita income features over a simple wealth ratio.

What we did not include. Q-Q plots test for normality, which is not assumed by any model in our pipeline (SVM, XGBoost, MLP, Random Forest). Testing for it adds no actionable information. Full pairplots with 10 engineered features produce 100 subplots, which is visually intractable and redundant with the correlation heatmap already present.

3. METHODOLOGY

3.1. Label Quality and Noise

Labels are only as reliable as the advisors who generated them. A conflicted advisor may produce labels that reflect distribution targets rather than genuine client needs, introducing structured noise into the training data. Rather than treating label noise as fully irreducible, we take two concrete steps to assess and mitigate its effect.

Noise-robust loss function. For XGBoost we replace the standard binary cross-entropy objective with Symmetric Cross-Entropy (SCE) [**Wang2019SCE**]:

$$\mathcal{L}_{\text{SCE}} = \alpha \cdot \text{CE}(y, \hat{p}) + \beta \cdot \text{CE}(\hat{p}, y)$$

where $\text{CE}(y, \hat{p}) = -y \log \hat{p}$ is the standard forward cross-entropy and $\text{CE}(\hat{p}, y) = -\hat{p} \log y$ is the reverse term. The reverse term penalizes overconfident predictions on potentially mislabeled samples, providing robustness to label noise without requiring knowledge of the noise rate. We treat α and β as hyperparameters tuned in the inner cross-validation loop.

Label sensitivity analysis. To quantify how much results depend on label integrity, we re-run the full pipeline after randomly flipping 5% and 10% of training labels independently, repeating each perturbation five times. If outer-fold F1 degrades by less than two percentage points at the 10% corruption level, we conclude the pipeline is label-robust.

3.2. Feature Sets and Ablation Design

Running all models on a single feature set conflates algorithmic contribution with feature contribution. We adopt a structured ablation: we compare the professor's baseline feature set against our theoretically motivated engineered set under identical evaluation conditions, using a Wilcoxon test on per-fold F1 scores.

Baseline feature set $\mathcal{F}_B$ (professor's set)

$$\mathcal{F}_B = \{\text{Age}, \ \text{Gender}, \ \text{FamilyMembers}, \ \text{FinEdu}, \ \text{RiskProp}, \ \log \text{Wealth}, \ \log \text{In}$$

This is the set used in the original notebook: raw variables plus `log1p` transforms for Wealth and Income. It serves as the ablation baseline.

Engineered feature set $\mathcal{F}_E$ (our redesign)

All features in $\mathcal{F}_E$ are motivated by financial theory first, then validated by their correlation with the targets and by the visual evidence from EDA.

$$\mathcal{F}_E = \{\text{Age}, \ \underbrace{\text{Age}^2}_{\text{nonlinear lifecycle}}, \ \text{FinEdu}, \ \text{RiskProp}, \ \log \text{Wealth}, \ \log \text{Income}, \ \underbrace{\text{Age} \times}_{\text{Age}}$$

where FM denotes FamilyMembers. The individual justifications are as follows.

Age^2 captures the non-linear acceleration of IncomeInvestment need after age 65, visible in the violin plots as a steeper divergence between need groups in the oldest cohort. A linear term alone cannot represent this curvature.

Age_x_Wealth = Age × log Wealth encodes the joint elderly-wealthy signal identified in the scatter plot analysis. Its correlation with the IncomeInvestment target is $r = 0.45$, substantially stronger than either Age ($r = 0.33$) or Wealth_log ($r = 0.40$) alone, confirming that the interaction captures information neither marginal variable can provide.

Income_per_FM_log = log(1 + Income/FM) and Wealth_per_FM_log = log(1 + Wealth/FM) replace raw Income_log and Wealth_log as the primary scale variables. Per-capita income and wealth are more informative than gross values for multi-member households: a household income of 80k has very different implications depending on whether it must support two or five people. Despite their construction from Income_log ($r = 0.91$) and Wealth_log ($r = 0.93$) respectively, these correlations are materially below 1.0, confirming that FamilyMembers introduces genuine additional signal that is not subsumed by the gross variables.

FinEdu_x_Risk = FinancialEducation × RiskPropensity captures the "sophisticated investor" profile: clients who are both financially literate and risk-tolerant. This interaction provides Logistic Regression with an explicit cross-product of the two collinear features ($r = 0.68$) it cannot reconstruct from correlated main effects alone.

Excluded variables. Gender has correlation $|r| < 0.015$ with both targets and is excluded as pure noise. Raw FamilyMembers is superseded by the per-capita ratio features; its gross value carries no residual information once the ratios are present. The Income/Wealth ratio

used in the original notebook is replaced by the per-capita features and the Age_x_Wealth interaction, which have stronger and more interpretable target correlations.

Multicollinearity is expected and not problematic. Age, Age^2, and Age_x_Wealth are correlated by construction. This is intentional. Tree-based models (Random Forest, XGBoost) are invariant to multicollinearity: split decisions depend only on the marginal distribution of each feature given the split point, not on inter-feature correlations. For Logistic Regression and SVM, L1 regularization and MinMaxScaler respectively mitigate the effect. The engineered feature correlation heatmap confirms the expected structure and rules out unexpected collinearities elsewhere in the set.

Ablation scope. The ablation is conducted for Logistic Regression (cannot learn interactions natively; $\mathcal{F}_E$ is expected to improve performance) and XGBoost (learns interactions natively; $\mathcal{F}_E$ is expected to be redundant). Naive Bayes is tested on $\mathcal{F}_B$ only: the Gaussian independence assumption is already violated by the base features, and adding constructed interactions compounds this violation without the benefit that motivates them for LR. All other models use $\mathcal{F}_E$. The winner of each pairwise ablation enters model selection as a single candidate.

3.3. Per-Model Preprocessing

A uniform preprocessing pipeline is statistically incorrect when applied across heterogeneous model families. We design separate pipelines matched to each model's inductive bias. All scalers are fit exclusively on the training fold; fitting on the full dataset before the split constitutes data leakage. Raw feature constructions (log transforms, products, ratios) involve no statistical fitting and are safe to apply before splitting.

Logistic Regression — StandardScaler. L1/L2 regularization shrinks coefficients with uniform strength λ. If features have different variances, the effective regularization varies across features at the same λ. StandardScaler (zero mean, unit variance) ensures the penalty is uniformly calibrated. L1 (`solver='liblinear'`) is used with $\mathcal{F}_E$ for automatic selection among collinear terms; L2 with $\mathcal{F}_B$.

Gaussian Naive Bayes — no scaling. GNB estimates per-class parameters $(\hat{\mu}_{kj}, \hat{\sigma}^2_{kj})$ for each feature independently. The likelihood ratio depends only on within-class distributions; rescaling changes $\hat{\mu}$ and $\hat{\sigma}$ proportionally, leaving the ratio invariant. GNB uses $\mathcal{F}_B$ only.

SVM (RBF) — MinMaxScaler to $[0,1]$. The RBF kernel $k(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ measures Euclidean distance. A feature in $[18, 97]$ contributes up to $(97-18)^2 = 6241$ to the squared distance; a feature in $[0,1]$ contributes at most 1. Without normalization the kernel is blind to small-scale features. `probability=True` enables `predict_proba` via Platt scaling, required for soft voting.

Random Forest and XGBoost — no scaling. A threshold split $x_j > t$ is invariant to any strictly monotonic transformation f: if $x_j > t$ then $f(x_j) > f(t)$. Linear scaling is a special case and has no effect on tree decisions. The log transforms of Wealth and Income matter because they change the *distribution* of split points, compressing the right tail and concentrating splits where observations are dense. XGBoost receives `scale_pos_weight = neg/pos` for the IncomeInvestment target only.

MLP — MinMaxScaler to $[0,1]$. Heterogeneous feature scales produce an elongated loss surface and slow gradient descent convergence. Batch Normalization normalizes layer activations, not input features, and does not substitute for input normalization. We use $[0,1]$ rather than zero-mean unit-variance to match the Sigmoid output activation.

Soft Voting Ensemble — Pipeline encapsulation. Each component is wrapped in a `sklearn.pipeline.Pipeline` with its own scaler, preventing leakage and ensuring correct transformation automatically during nested cross-validation.

3.4. Evaluation Framework and MiFID II Constraints

The regulatory context imposes an explicit, asymmetric cost structure on misclassification. A false positive (predicting need = 1 when need = 0) results in recommending an unsuitable product — a potential MiFID II mis-selling event with regulatory and reputational consequences. A false negative (predicting need = 0 when need = 1) results in a missed sales opportunity — a revenue cost but no legal exposure.

Primary metric: F1-score. F1 balances precision and recall symmetrically, preventing degenerate solutions that maximize one at the expense of the other.

Hard constraint: Precision $\geq$ 0.75. At least three-quarters of positive predictions must correspond to genuine needs before a recommendation is considered safe to automate. This directly encodes the MiFID II compliance requirement.

Threshold selection via Precision–Recall curve. Fixing the classification threshold at 0.5 is arbitrary and ignores the cost asymmetry. For each model we plot the full Precision–Recall curve and select the operating threshold as the point that maximizes F1 subject to Precision ≥ 0.75. This threshold is a business decision, not a modeling convention, and varies across models and targets.

3.5. Nested Cross-Validation

A standard k-fold procedure that evaluates a fixed hyperparameter configuration confounds model selection with hyperparameter optimization, introducing selection bias. We use **nested cross-validation**: an outer 10-fold loop for model comparison and final performance estimation, and an inner 3-fold loop for hyperparameter tuning within each outer training fold. This produces unbiased performance estimates for each hyperparameter-optimized model. The inner loop uses random search over a predefined hyperparameter grid.

Final selection between top candidates uses a **Wilcoxon signed-rank test** on the ten outer-fold F1 scores. We use 10 outer folds rather than 5 specifically to ensure adequate statistical power: with only 5 paired observations the minimum achievable p-value is approximately 0.063, making it structurally impossible to reject the null at the 5% level regardless of the observed differences [**Demsar2006**]. If $p > 0.05$, Occam's razor selects the simpler model.

3.6. Calibration

A model with F1 = 0.78 and well-calibrated probabilities is more valuable for this pipeline than a model with F1 = 0.80 and poorly calibrated ones, because the recommendation engine uses `predict_proba` outputs as propensity *scores* to rank clients. A score of 0.70 must mean that approximately 70% of clients with that score actually hold the need; otherwise the confidence segmentation and priority ranking are meaningless.

We assess calibration both visually via **reliability diagrams** and quantitatively via the **Brier score** [**Brier1950**]:

$$\mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n}(\hat{p}_i - y_i)^2$$

The Brier score is a proper scoring rule: lower is strictly better, and 0.25 is the no-skill baseline for a balanced binary target. We report it pre- and post-calibration to quantify the improvement from isotonic regression. Isotonic regression is preferred over Platt scaling because it is nonparametric and does not assume a sigmoidal calibration error form [**Niculescu2005**].

3.7. Model Candidates

Tier 1 — Interpretable baselines. *Logistic Regression* (L1 and L2) is the theoretically motivated model: the dominance of Age and Wealth_log as predictors of IncomeInvestment suggests approximate

linear separability in log-transformed space. Coefficients are interpretable as log-odds ratios, satisfying MiFID II auditability requirements. Probabilities are naturally calibrated. *Gaussian Naive Bayes* is the generative baseline, modeling $P(\mathbf{x} \mid y)$ and inverting via Bayes' theorem.

Tier 2 — Nonlinear models (with ablation). *Random Forest* applies bootstrap aggregation over decision trees, reducing variance by averaging decorrelated predictors. *XGBoost* applies sequential gradient boosting, directly minimizing log-loss stage-wise with explicit regularization and a noise-robust SCE objective. Both are post-hoc calibrated via isotonic regression.

Tier 3 — Neural baseline. A Multilayer Perceptron with architecture $16 \to 8 \to 1$ (BatchNorm, Dropout = 0.2, ReLU, Sigmoid output, BCE loss, Adam, ReduceLROnPlateau). The architecture is deliberately compact: with approximately 4,000 training observations and 10 input features, the $64 \to 32 \to 16 \to 1$ architecture used in preliminary experiments contains ∼3,000 parameters — overparameterized for this dataset and consistently outperformed by tree-based methods, consistent with recent empirical evidence [**Grinsztajn2022**].

Tier 4 — Soft Voting Ensemble. A soft voting classifier averaging calibrated probabilities from Logistic Regression (best ablation variant), XGBoost (best ablation variant), and calibrated SVM. The three components fail in structurally different regimes: LR fails when the decision boundary is nonlinear; XGBoost is sensitive to distribution shift and outliers; SVM fails when classes are not well-separated in the RBF kernel space. Because their errors arise from different mechanisms they are partially uncorrelated; averaging calibrated probabilities reduces total generalization error relative to any individual component. Soft voting is preferable to hard voting because it uses the full probability information before discretization, producing better-calibrated aggregate predictions.

A **Classifier Chain** (XGBoost base estimator) is additionally tested. AccumulationInvestment is predicted first, and its prediction is passed as a feature to the IncomeInvestment model. The ordering follows the life-cycle hypothesis [**Modigliani1954**]: accumulation is the earlier life stage, so conditioning the income-need model on whether accumulation need is present encodes the directional financial prior that retirement-oriented needs succeed working-age ones. Although the empirical inter-target correlation is near zero ($r = 0.011$), the chain encodes this structural prior and is compared empirically.

3.8. Recommendation Pipeline Design

Stage 1 — Propensity scoring. The winning model generates calibrated propensity scores $\hat{p}(y = 1 \mid \mathbf{x})$ for all 5,000 clients on *both* need types. This replaces binary classification: a score of 0.88 and a score of 0.53 both exceed the 0.5 threshold but carry fundamentally different information about confidence. Clients are segmented as: high confidence ($\hat{p} > 0.75$, automated recommendation), uncertain ($\hat{p} \in [0.40, 0.75]$, routed to human advisor with SHAP force-plot explanation), and low propensity ($\hat{p} < 0.40$, no action). The uncertain segment is the operational MiFID II safeguard: no automated decision is made when the model lacks confidence, preserving human judgment where it is most needed.

Stage 2 — Product matching. For each high-confidence client, products of the matching type are scored using a suitability function derived from mean-variance utility [**Markowitz1952**]. Under mean-variance preferences, utility is maximized at the portfolio risk level equal to the client's risk propensity; deviating from this optimum reduces expected utility approximately linearly in the absolute deviation:

$$s(c, p) = \begin{cases} 1 - \left|\text{RiskPropensity}_c - \text{Risk}_p\right| & \text{if Risk}_p \le \text{RiskPropensity}_c \\ 0 & \text{if Risk}_p > \text{RiskPropensity}_c \end{cases}$$

Setting $s = 0$ for over-risk products enforces the hard MiFID II constraint. The function is symmetric: excessive conservatism is penalized equally to equivalent over-risk, reflecting the principle that suboptimal conservatism also harms client outcomes. Both risk quantities are normalized to $[0, 1]$ and $[0.12, 0.88]$ respectively, making the absolute deviation directly comparable across clients. The function is applied to both need types independently.

Stage 3 — Priority ranking. The advisor action list is ranked by:

$$\text{Priority}(c, p) = \hat{p}(y = 1 \mid \mathbf{x}_c) \cdot s(c, p)$$

This ranks clients by the joint probability of having the need and receiving a well-matched product, telling the advisor not only *who* to contact but in *what order*.

SVD-based collaborative filter. As a complementary discovery layer, we implement a matrix factorization recommender based on SVD [**Koren2009**]. The input is the $5{,}000 \times 11$ implicit feedback matrix $R \in \{0, 1\}^{n \times m}$, where $R_{cp} = 1$ if client c purchased product p. We factorize $R \approx U \Sigma V^\top$ retaining the top-k singular components (with k selected via held-out reconstruction error) and score unseen pairs as $\hat{r}_{cp} = \mathbf{u}_c^\top \mathbf{v}_p$. Products surfaced by SVD are still subject to the hard risk cap and must belong to the client's predicted need type before entering the final ranked list. The two approaches are complementary: SVD for discovery, propensity-constrained matching for compliance confirmation.

4. RESULTS

4.1. Feature Engineering Ablation

Table 2. Feature engineering ablation: nested CV outer-fold F1 (mean ± std). $\mathcal{F}_B$ = professor's baseline; $\mathcal{F}_E$ = our engineered set.

Model	Features	Accum. F1	Income F1
LR (L1)	$\mathcal{F}_B$	±	±
LR (L1)	$\mathcal{F}_E$	±	±
XGBoost	$\mathcal{F}_B$	±	±
XGBoost	$\mathcal{F}_E$	±	±

For Logistic Regression, $\mathcal{F}_E$ yields a statistically significant improvement (Wilcoxon p for both targets). The gain is attributable primarily to `FinEdu_x_Risk`, which provides LR with an explicit cross-product of the two collinear features it cannot reconstruct from main effects alone, and secondarily to `Age_x_Wealth`, which concentrates the elderly-wealthy interaction signal in a single coefficient.

For XGBoost, the difference between $\mathcal{F}_B$ and $\mathcal{F}_E$ is not statistically significant (p for both targets) and $\mathcal{F}_B$ marginally dominates in point estimates. This confirms the theoretical argument: XGBoost learns all relevant interactions natively through sequential splits. Adding pre-computed interaction terms introduces correlated features that dilute SHAP-based importance attribution without improving predictive performance.

Ablation conclusion: LR enters model selection trained on $\mathcal{F}_E$; all other models use $\mathcal{F}_E$ as well (except Naive Bayes, which uses $\mathcal{F}_B$).

4.2. Model Selection

Table 3. Nested CV outer-fold F1 (mean ± std) with optimized hyperparameters.

Model	Accum. F1	Income F1
LR (L1, $\mathcal{F}_E$)	±	±
Naive Bayes ($\mathcal{F}_B$)	±	±
Random Forest (cal., $\mathcal{F}_E$)	±	±
XGBoost (cal., $\mathcal{F}_B$)	±	±
MLP 16→8→1 ($\mathcal{F}_E$)	±	±
Classifier Chain ($\mathcal{F}_E$)	±	±
Soft Voting Ensemble	±	±

The Soft Voting Ensemble achieves the highest mean F1 on both targets with the lowest fold-to-fold variance. A Wilcoxon test between the Ensemble and XGBoost yields p for AccumulationInvestment and p for IncomeInvestment. For the Income target the difference is not significant at the 5% level; we retain the Ensemble for both tasks on the basis of lower variance, architectural consistency, and the theoretical argument that base-model diversity reduces error when targets are partially uncorrelated.

Logistic Regression performs strongly (ΔF1 $\approx$ behind the ensemble), corroborating the EDA finding that Age and Wealth_log dominate the signal for IncomeInvestment: if the true boundary is approximately linear in these features, LR captures most of the available signal and gains little from ensemble complexity. The MLP underperforms tree-based models on both targets, consistent with the parameters-to-samples argument. The Classifier Chain matches the Ensemble in point estimate for IncomeInvestment but exhibits marginally higher variance, consistent with the near-zero inter-target correlation limiting the benefit of the chain.

4.3. Threshold Selection

For each model we plot the full Precision–Recall curve and select the operating threshold as the point that maximizes F1 subject to Precision ≥ 0.75. At the chosen threshold, the Ensemble achieves Precision = and Recall = for AccumulationInvestment, and Precision = and Recall = for IncomeInvestment. The selected thresholds are and respectively, both below the conventional 0.5 — reflecting the class imbalance in the Income target and confirming that fixed-threshold evaluation under-reports true model utility.

4.4. Calibration

Reliability diagrams confirm that pre-calibration Random Forest scores are biased toward 0.5. Post-calibration with isotonic regression corrects this to near-diagonal. XGBoost is well-calibrated for AccumulationInvestment and overconfident for IncomeInvestment at high predicted probabilities. The Soft Voting Ensemble, combining three calibrated components with distinct error patterns, produces the most consistent reliability across the full probability range. Table 4 reports Brier scores pre- and post-calibration; isotonic regression consistently reduces the score below the no-skill baseline of 0.25, with the Ensemble achieving the lowest post-calibration value on both targets.

Table 4. Brier scores pre- and post-calibration (lower is better; 0.25 = no-skill baseline).

	Accum. BS		Income BS	
Model	Pre	Post	Pre	Post
Random Forest				
XGBoost				
Soft Voting Ensemble				

4.5. Label Sensitivity Analysis

Table 5 reports outer-fold F1 of the Soft Voting Ensemble under random label corruption, averaged over five independent perturbations per level. A degradation below two F1 points at the 10% corruption level confirms robustness of the revealed-preference labeling assumption.

4.6. Explainability

SHAP analysis (TreeExplainer on the XGBoost component) identifies Wealth_log and Age as the dominant features for IncomeInvestment, with positive SHAP values concentrated among high-wealth, older clients — a direct empirical confirmation of the life-cycle hypothesis. For AccumulationInvestment, RiskPropensity and Income_log lead the ranking. `Age_x_Wealth` ranks among the top features for IncomeInvestment, validating the scatter plot motivation: the joint elderly-wealthy signal is genuinely informative beyond the marginal predictors. The alignment between SHAP rankings and EDA-observed marginal correlations provides a coherence check that the model recovers the structure visible in the raw data, now in a nonlinear, interaction-aware form.

Table 5. Label sensitivity: mean F1 under random label corruption (5 repetitions per level).

Target	0% (baseline)	5% flipped	10% flipped
AccumulationInvestment			
IncomeInvestment			

4.7. Recommendation Results

On the full client dataset, the pipeline achieves % coverage for AccumulationInvestment clients and % for IncomeInvestment clients at the chosen confidence threshold. The remaining clients are routed to human advisors. The suitability pass rate ($\text{Risk}_p \leq \text{RiskPropensity}_c$) is 100% by construction of the hard constraint for both need types.

The SVD-based recommender achieves % coverage by leveraging latent purchase patterns but produces a suitability pass rate of % without explicit risk enforcement. The two approaches are complementary: SVD for discovery, classification for compliance-constrained confirmation.

5. DISCUSSION

5.1. XGBoost Alone Is Insufficient

XGBoost is a discriminative, frequentist model producing a single point estimate: "given these features, the predicted probability of need is 0.73." This estimate carries no uncertainty quantification. Two clients with predicted probabilities of 0.51 and 0.95 are both classified as positive under hard thresholding but carry fundamentally different implications for the recommendation pipeline. The Soft Voting Ensemble mitigates this: disagreement among the three base models implicitly encodes uncertainty, flagging ambiguous clients for human review without requiring explicit Bayesian inference. Furthermore, XGBoost's `predict_proba` output is not a true posterior probability without calibration. For a compliance-sensitive system where probabilities directly determine whether a recommendation is automated or escalated, calibration is not optional.

5.2. The Case Against a Uniform Preprocessing Pipeline

Applying a uniform MinMaxScaler to all models is inconsistent with each algorithm's statistical assumptions. For Logistic Regression with L2 regularization, MinMaxScaler does not equalize variance: two features with different within-range distributions receive incomparable effective regularization at the same λ. StandardScaler equalizes both mean and variance, ensuring uniform penalty calibration. For tree models, any scaler is strictly equivalent to no scaler. For the MLP, $[0, 1]$ input matches the Sigmoid output activation. A single pipeline can be correct for at most one model family simultaneously.

5.3. Implications of Near-Zero Inter-Target Correlation

The near-zero correlation ($r = 0.011$) between targets means that a client predicted to have Accumulation need carries almost no information about their Income need. The recommendation system evaluates both needs independently for every client, and a client may receive products of both types if both propensity scores exceed the confidence

threshold. This is not an oversight but the statistically correct behavior given the observed joint distribution.

6. CONCLUSION

We present an end-to-end investment needs estimation and recommendation system grounded in both statistical evidence from the data and theoretical priors from financial economics. The organizing principle throughout is regulatory compliance: every design choice from metric selection to threshold calibration to product matching enforces or operationalizes a MiFID II requirement.

Key findings: (1) AccumulationInvestment and IncomeInvestment are empirically near-independent ($r = 0.011$), contradicting the prior of mutual exclusivity and justifying independent binary classifiers; (2) IncomeInvestment is the more predictable target, dominated by Age and Wealth_log in direct confirmation of the Modigliani life-cycle hypothesis; (3) feature engineering provides significant lift for Logistic Regression but is statistically redundant for XGBoost, demonstrating that the benefit of hand-crafted interactions is contingent on whether the model can learn them natively; (4) per-model preprocessing is necessary for statistical correctness; (5) calibrated propensity scores are essential inputs to the recommendation engine — binary classifications are insufficient; (6) operating threshold selection from the PR curve under Precision ≥ 0.75 is more honest than a fixed threshold of 0.5; (7) the pipeline is robust to moderate label noise as confirmed by sensitivity analysis.

7. SOURCE AND REPRODUCIBILITY

All modeling pipelines, ablation procedures, calibration analyses, and recommendation engine implementations were developed entirely by the authors. The complete source code is hosted on GitHub at this link.